

Advanced Natural Language Processing

Discussion Text Representation

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Learning Objectives

- Explain why computers need numeric representations of text, and what is lost in the process
- Compute and interpret TF-IDF weights and cosine similarity between documents
- Explain why bag-of-words / vector-space models cannot capture word order
- Contrast static word embeddings (Word2Vec) with contextual embeddings (BERT), and explain why the latter solve the polysemy problem
- Explain how BERT's WordPiece tokenizer handles out-of-vocabulary words via subwords

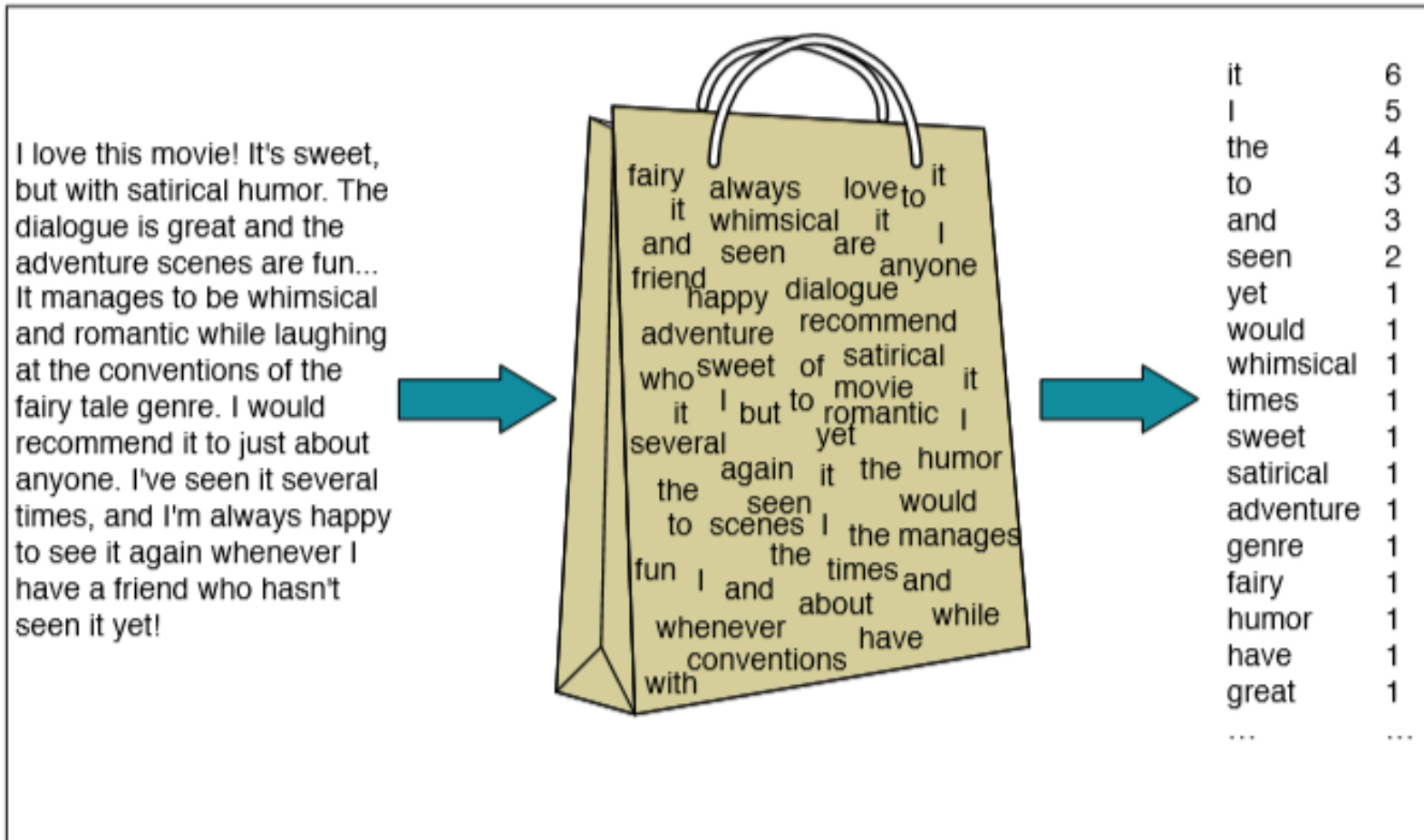


How do computer deal with text?

- What happens exactly when someone uses a computer program to deal with natural language? Computers are digital machines with limited capabilities.
- Computers cope best with numbers, in particular whole numbers, also known as integers, if it has to be really fast.
- People are analogue, and human language is also analogue, full of dualities, ambiguity, inconsistencies, interference, errors and exceptions.
- If we search for something then we often think in concepts, senses and meanings, all areas in which a computer cannot directly deal with.



Bag-of-Words Representation



Teaching note: Each document becomes a vector of word counts — order is discarded entirely (see slide 6).

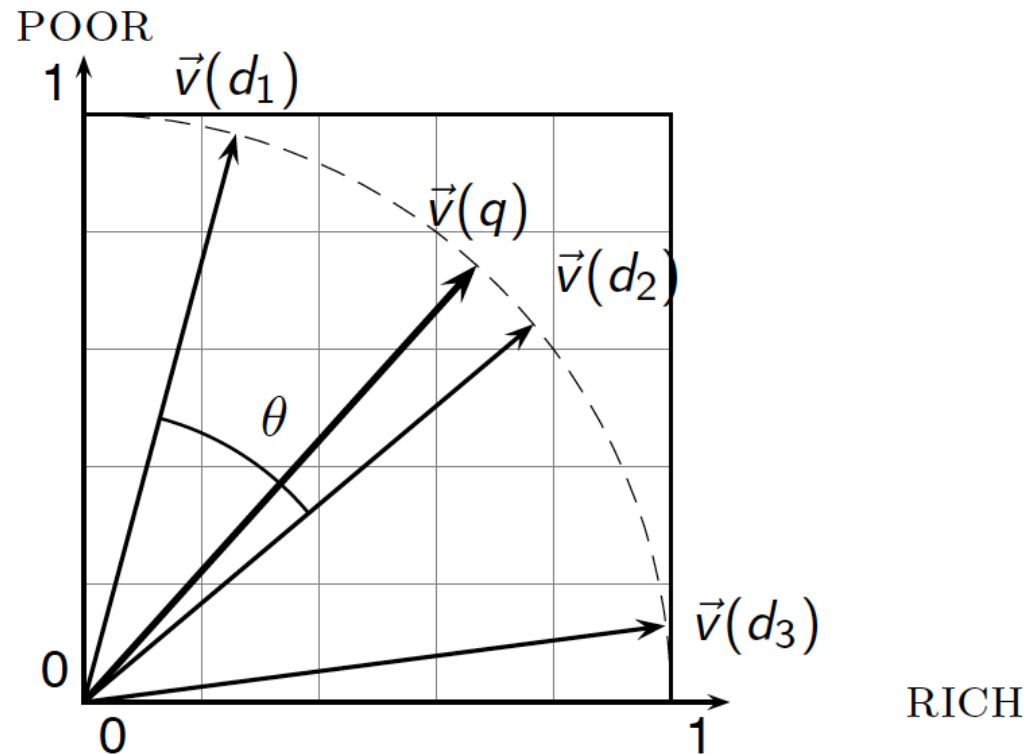
Binary → count → weight matrix

Teaching note: $\text{tf-idf}(t,d) = \text{tf}(t,d) \times \log(N / \text{df}(t))$ — rows are terms, columns are documents; high weight = frequent here, rare elsewhere.

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	5.25	3.18	0	0	0	0.35
Brutus	1.21	6.1	0	1	0	0
Caesar	8.59	2.54	0	1.51	0.25	0
Calpurnia	0	1.54	0	0	0	0
Cleopatra	2.85	0	0	0	0	0
mercy	1.51	0	1.9	0.12	5.25	0.88
worser	1.37	0	0.11	4.15	0.25	1.95

Each document is now represented by a real-valued vector of tf-idf weights $\in \mathbb{R}^{|V|}$

Documents as vectors – Document Collections as Matrices: similarity by measuring angles / cosines.



Teaching note: $\cos(\theta) = (\mathbf{A} \cdot \mathbf{B}) / (\|\mathbf{A}\| \|\mathbf{B}\|)$ — measures orientation, not magnitude, so document length doesn't skew similarity.

How about word order?

- Lost with both TF-IDF and with Cosine distance.
- “Jane hit Jon” same vector / angle as “Jon hit Jane”

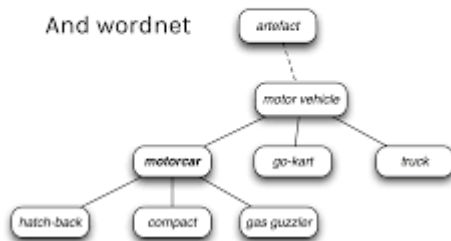


Thesaurus or Ontology: Wordnet

- Meaning
- Associated words (opposite, similar meaning)



And wordnet



<https://wordnet.princeton.edu/>

Teaching note: Word embeddings (next slide) solve exactly this: they replace random hash-lookup addresses with vectors where distance reflects meaning.

Hamburger

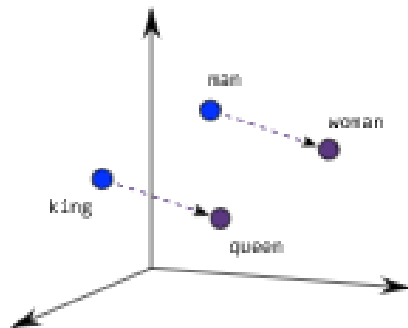
- Hamburger (an inhabitant of Hamburg)
 - direct hypernym:
 - German (a person of German nationality)
 - sister term
 - German (a person of German nationality)
 - East German (a native/inhabitant of the former GDR)
 - Bavarian (a native/inhabitant of Bavaria)
- derivationally related form
 - Hamburg (a port city in northern Germany on the Elbe River that was founded by Chalemagne in the...)

The Problem:

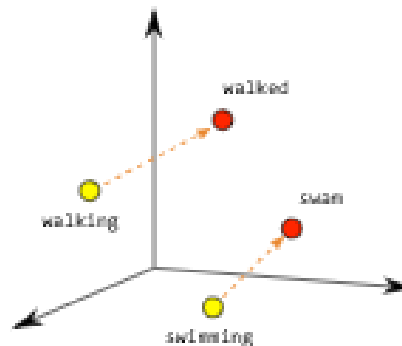
- In traditional NLP systems, tokens are represented by hash-values, which can be used as lookup address.
- Hash values are random and have a flat distribution.
- No relation to a linguistic meaning.
- No problem in WordNet or rules-based approaches, but when used in deep-learning models: no relation between encoding and meaning.



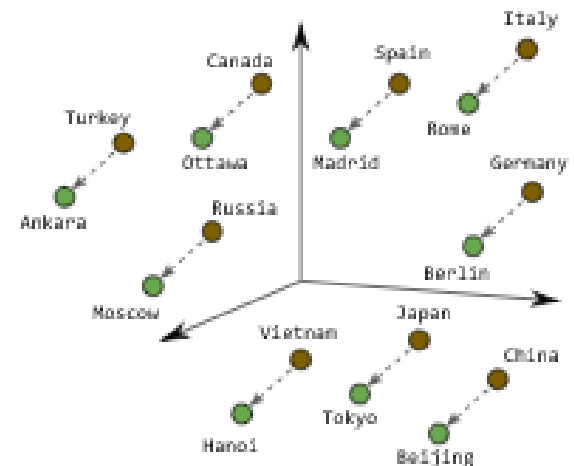
Word Embeddings: similar linguistic context, similar numeric representations



Male-Female



Verb Tense



Country-Capital

Teaching note: Classic analogy: $\text{vec}(\text{"king"}) - \text{vec}(\text{"man"}) + \text{vec}(\text{"woman"}) \approx \text{vec}(\text{"queen"})$.

Each panel shows one semantic axis learned purely from co-occurrence: gender, verb tense, capital-country.

Why Word Embeddings?

- Pre-trained language model.
- Understand context better.
- Transfer learning: understand already general aspects of language, subsequent only need to fine-tune for a specific NLP task.
- No need for millions or billions of annotated training data (when using deep learning) when learning a specific linguistic task.



This goes back to Wittgenstein

In 1922 Ludwig Wittgenstein published the **Tractatus Logico-Philosophicus**.

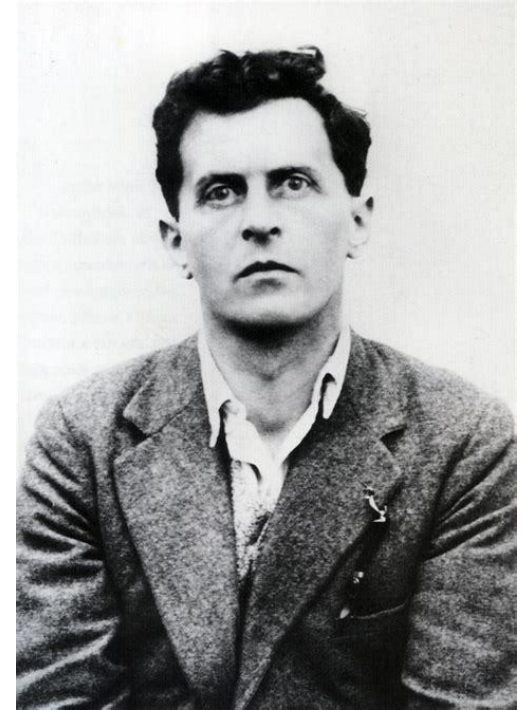
The project had a broad goal: to identify the relationship between language and reality and to define the limits of science.

He discusses the process of identifying a word's meaning from the context of its usage.

It can be argued that the Tractarian thoughts can be understood as acts of geometric projection in logical space, performed by the transcendental subject.

See also:

- Wittgenstein, Ludwig. 1953. *Philosophical Investigations*. John Wiley & Sons
- Tractatus Logico-Philosophicus, Logisch-philosophische Abhandlung By Ludwig Wittgenstein, First published by Kegan Paul (London), 1922.
- Projective Geometry in Logical Space: Rethinking Tractarian Thoughts Article in International Journal of Philosophical Studies · December 2017 DOI: 10.1080/09672559.2017.1404623 by Pablo Acuna.



Teaching note: Note: the source usually cited in NLP for “meaning from context” is J.R. Firth's distributional hypothesis (1957): “you shall know a word by the company it keeps.”

Let's Start understanding Word2Vec (2013)

- We're going to train a simple (back propagation) neural network with a single hidden layer to perform a certain task, but then we're not actually going to use that neural network for the task we trained it on! Instead, the goal is actually just to learn the weights of the hidden layer—we'll see that these weights are actually the “word vectors” that we're trying to learn.
- Given a specific word in the middle of a sentence (the input word), look at the words nearby and pick one at random. The network is going to tell us the probability for every word in our vocabulary of being the “nearby word” that we chose.
- We will call this Skip-grams: a technique where by n-grams are still stored to model language, but they allow for tokens to be skipped to overcome the data sparsity problem. We analyze this by computing all possible skip-grams in a training corpus and measure how many adjacent (standard) n-grams these cover in test documents.

See also: Thomas Mikolov et al., Efficient Estimation of Word Representations in Vector Space (arXiv:1301.3781, January 2013), and Distributed Representations of Words and Phrases and their Compositionality (NIPS 2013)



Word2Vec: Two Architectures

- **Skip-gram:** given the centre word, predict the surrounding context words. Works well with small data and rare words.
- **CBOW (Continuous Bag-of-Words):** given the surrounding context words, predict the centre word. Faster to train, works well with frequent words.
- **Both:** learn the same kind of hidden-layer weights — those weights become the word vectors, regardless of which direction the prediction runs.

Takeaway: the direction of prediction differs, but both architectures rest on the same distributional hypothesis: similar context → similar vector.



Bias in Word2Vec

$$\vec{\text{man}} - \vec{\text{woman}} \approx \vec{\text{king}} - \vec{\text{queen}}$$

$$\vec{\text{man}} - \vec{\text{woman}} \approx \vec{\text{computer programmer}} - \vec{\text{homemaker}}.$$

Extreme *she* occupations

- | | | |
|-----------------|-----------------------|------------------------|
| 1. homemaker | 2. nurse | 3. receptionist |
| 4. librarian | 5. socialite | 6. hairdresser |
| 7. nanny | 8. bookkeeper | 9. stylist |
| 10. housekeeper | 11. interior designer | 12. guidance counselor |

Extreme *he* occupations

- | | | |
|----------------|-------------------|----------------|
| 1. maestro | 2. skipper | 3. protege |
| 4. philosopher | 5. captain | 6. architect |
| 7. financier | 8. warrior | 9. broadcaster |
| 10. magician | 11. fighter pilot | 12. boss |

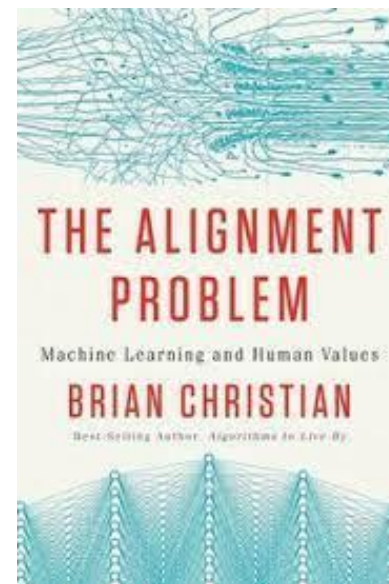
Teaching note: Bias arises because vectors encode corpus co-occurrence statistics — if text co-occurs “doctor” with “he” more than “she,” the geometry reflects that. See the ethics/fairness lecture.

They typed:

doctor - man + woman

The answer came back:

nurse



We will discuss this in more detail in the tutorial

BERT (Sub-)words

Internally, BERT stores 3 types of (sub)words:

- 1) “Context-free” pre-trained embeddings
- 2) “Context-based” pre-trained embeddings
- 3) “Context-averaged” pre-training embeddings

Sub-words:

The original BERT vocabulary is fixed at 30.000 tokens. Words that are not part of the core vocabulary are represented as subwords and characters: notice how the word “embeddings” can be represented: ['em', '##bed', '##ding', '##s']

The WordPiece Tokenizer

- BERT uses what is called a WordPiece tokenizer.
 - Words are tokenized either into the full forms if it is a known word (part of a 30k vocabulary, then one word becomes one token), or
 - Words are tokenized into word pieces (if it is not part of the 30k vocabulary), then one word can be broken into multiple tokens.

We will discuss this in more detail in the tutorial

Teaching note: Why it matters: any word can be built from known sub-word pieces, so the model never hits a true “unknown word” wall.

Word	Token(s)
surf	['surf']
surfing	['surf', '##ing']
surfboarding	['surf', '##board', '##ing']
surfboard	['surf', '##board']
snowboard	['snow', '##board']
snowboarding	['snow', '##board', '##ing']
snow	['snow']
snowing	['snow', '##ing']

Context Sensitive Embeddings

Sentences

"bank"
"he eventually sold the shares back to the bank at a premium."
"the bank strongly resisted cutting interest rates."
"the bank will supply and buy back foreign currency."
"the bank is pressing us for repayment of the loan."
"the bank left its lending rates unchanged."
"the river flowed over the bank."
"tall, luxuriant plants grew along the river bank."
"his soldiers were arrayed along the river bank."
"wild flowers adorned the river bank."
"two fox cubs romped playfully on the river bank."
"the jewels were kept in a bank vault."
"you can stow your jewellery away in the bank."
"most of the money was in storage in bank vaults."
"the diamonds are shut away in a bank vault somewhere."
"thieves broke into the bank vault."
"can I bank on your support?"
"you can bank on him to hand you a reasonable bill for your services."
"don't bank on your friends to help you out of trouble."
"you can bank on me when you need money."
"i bank on your help."

The different meanings of the word "bank" will get very different vectors, depending on this context. So, the "financial bank" will be closer to other financial words than the "river bank". This is essential for good language understanding!

Teaching note: This is the direct fix for both WordNet's hash-lookup problem (slide 7) and Word2Vec's one-vector-per-word limit (slide 8).

mouse (rodent)

mouse (computer)

So, we need different vectors (directions) for different meanings.

Now let's form groups of 4



Discussion Lecture: Text Representation



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Teaching note: Use the ideas from the lecture (TF-IDF, embeddings, BERT) as ammunition — there's no single correct answer to these.

Questions on the lecture?

Discuss Exercises Lecture on Text Representation

Exercise 1

Consider a collection of $N = 10,000$ documents. You are analyzing two terms:

- Term A: $df_A = 10$ (appears in 10 documents)
- Term B: $df_B = 1,000$ (appears in 1,000 documents)
(Use $\text{idf}_t = \log_{10}(N/df_t)$ and log-frequency term weighting $w_{t,d} = 1 + \log_{10}(\text{tf}_{t,d})$ for $\text{tf}_{t,d} > 0$, matching the lecture slides.)
- Calculate the idfweights for both Term A and Term B. If Document d_1 contains Term A with a raw frequency count of $\text{tf}_{A,d_1} = 100$, what is the final tf-idfweight of Term A in document d_1 ?
- Suppose document d_2 is created by duplicating all the contents of d_1 (so every word occurs exactly twice as often).
 - Explain why the Euclidean distance between vectors $\vec{v}(d_1)$ and $\vec{v}(d_2)$ is greater than 0.
 - What is the cosine similarity $\cos(\theta)$ between $\vec{v}(d_1)$ and $\vec{v}(d_2)$, and why does this make angle/cosine comparison superior to Euclidean distance for document retrieval?
- Why does the Bag of Words (BoW) assumption fail to distinguish between "*Jane hit Jon*" and "*Jon hit Jane*", and why are raw document vectors in $\mathbb{R}^{|V|}$ extremely sparse (often $< 0.01\%$ non-zero)?

Exercise 2

Word2Vec represents words as dense continuous vectors rather than sparse, high-dimensional one-hot vectors.

a. Given the training sentence:

- "The quick brown fox jumps over the lazy dog."
- List all positive training sample pairs (center_word, context_word) generated when the sliding window selects "**brown**" as the target center word with a context window radius of 2 words on either side.

b. In the Skip-gram architecture shown on slide 34, suppose the vocabulary size is $|V| = 10,000$ and the hidden linear layer has $N = 300$ neurons:

- What are the dimensions of the hidden layer weight matrix?
- Explain how this weight matrix directly acts as a lookup table for the dense 300-dimensional word embeddings.

c. One key discovery of Word2Vec was semantic compositionality via vector arithmetic (e.g., $\vec{v}_{\text{king}} - \vec{v}_{\text{man}} + \vec{v}_{\text{woman}} \approx \vec{v}_{\text{queen}}$). What fundamental limitation prevented static Word2Vec models from disambiguating homonyms like "*river bank*" versus "*money bank*"?

Exercise 3

Compare the architectural differences between static word embeddings (Word2Vec) and contextualized transformer embeddings (BERT).

- Look at the embedding dimensions table on slide 42. What is the dimensionality of classic Word2Vec versus BERT-Base and modern large models (e.g., Llama 3 or text-embedding-3-large)? Why does modern NLP move from 300 dimensions up to several thousands?
- Slide 45 outlines that the input representation in BERT is the element-wise sum of three distinct embedding vectors. Name these three components and briefly state what information each component provides to the model.
- In a downstream task containing the sentences:
 - *"The river flowed over the bank."*
 - *"The bank left its lending rates unchanged."*
 - Explain how the resulting vector representation of the token "bank" differs between Word2Vec and BERT, and describe how self-attention enables BERT to separate the vectors geometrically in latent space.

Discussion Lecture: Text Representation



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Teaching note: Use the ideas from the lecture (TF-IDF, cosine, Word2Vec, BERT) as ammunition — there's no single correct answer to these.

On Geometry and Meaning: Can a Meaning Be a Place?

Every representation in this lecture, from tf-idf to BERT, rests on one assumption: that a meaning can be a location in a space, and that similar meanings are nearby. Distance stands in for similarity.

But “good” and “bad” occur in the same contexts and end up close together; “not” is near everything; and no point in the space can say that something is false. Which aspects of meaning can geometry express, and which can it not express even in principle?

When we accept cosine similarity as a measure of “meaning the same”, what notion of meaning have we silently agreed to?

On Word2Vec Compositionality: True Reasoning or Statistical Mirage?

One of the most famous results of Word2Vec is linear compositionality: vector arithmetic seemingly solves analogies such as $v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$.

Did the network discover an abstract conceptual ontology of gender and monarchy, or did it exploit statistical regularities in how words co-occur within a small sliding window?

The same arithmetic also yields $v(\text{doctor}) - v(\text{man}) + v(\text{woman}) \approx v(\text{nurse})$. Is the model reasoning, reflecting, or merely remembering? And if it only remembers, who is responsible for what it remembers?

On Contextualization and the Polysemy Paradox

BERT gives the word “bank” a different vector in every sentence it occurs in. There is no longer one point that is “bank”; there are only the contexts.

Wittgenstein argued that the meaning of a word is its use (slide 30). BERT seems to implement exactly that. But if meaning is use, what is the thing that is being used? Is there still a word underneath the contexts, or only the contexts?

If two very different vectors in two texts are both “bank”, what makes them the same word? And does the model need to know that at all, or only we?



Tutorial 02: Text Representation



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Questions?



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Teaching note: Recap: text $\rightarrow$ counts (BOW/TF-IDF) $\rightarrow$ static vectors (Word2Vec) $\rightarrow$ contextual vectors (BERT) — each step recovers a bit more meaning.